

MedDiagnosis

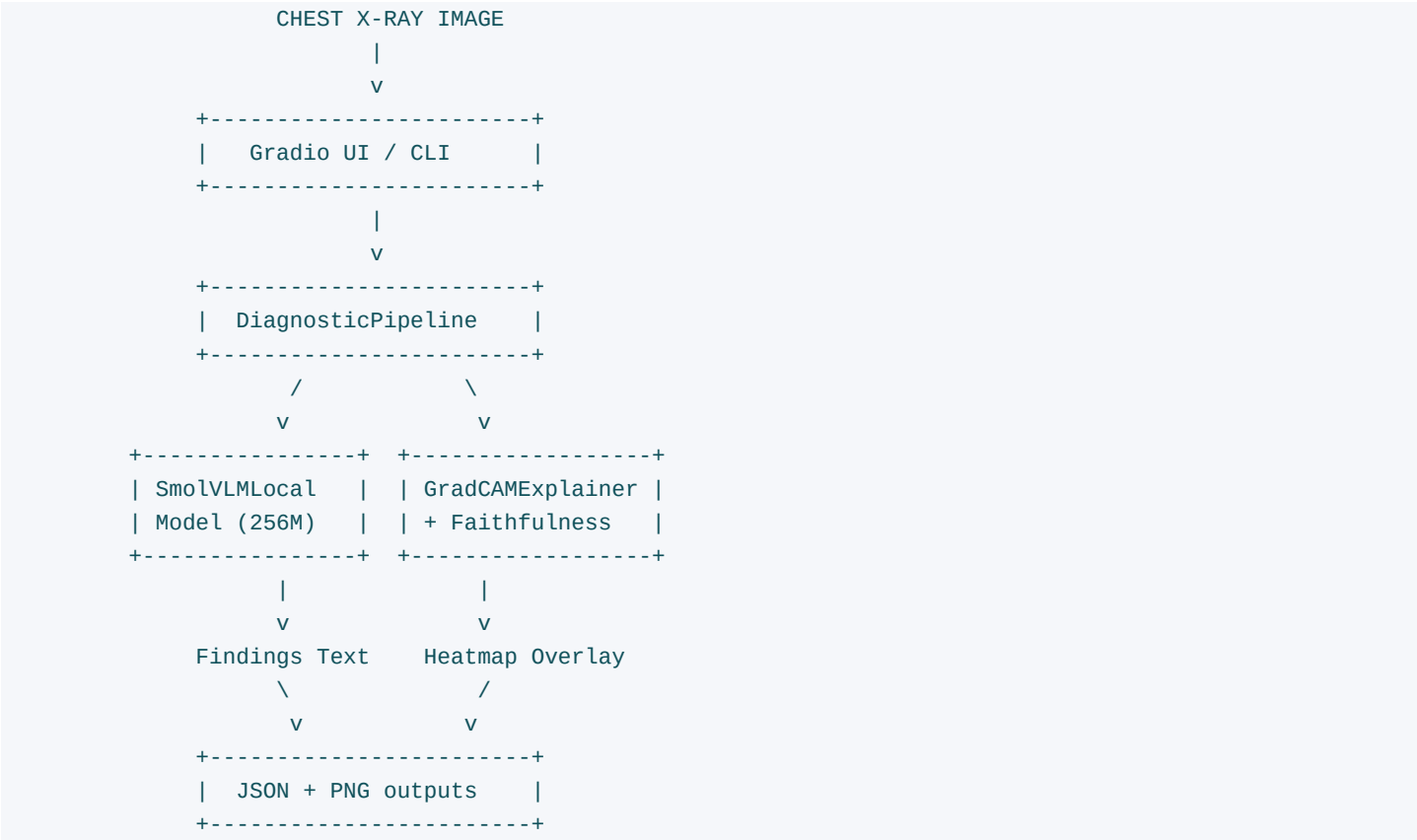
Explainable AI Framework for Automated Chest X-Ray Diagnosis

Version 0.1.0 | Research & educational use only - not for clinical diagnosis.

1. Project Overview

MedDiagnosis is an M.Tech research prototype that combines a lightweight biomedical Vision-Language Model (VLM) with an explainability layer for chest X-ray analysis. The system accepts a radiograph and a clinical prompt, generates findings and diagnosis text, optionally produces visual attribution maps (Grad-CAM / input-gradient saliency), and evaluates whether the model's explanation aligns with where it visually focused (faithfulness grounding). A Gradio web interface supports interactive analysis and follow-up chat on generated reports.

2. System Architecture



2.1 Data Flow

1. User uploads chest X-ray via CLI or Gradio.
2. Image and prompt are tokenized through SmolVLM processor (chat template).
3. BMC-smolVlm1-256M generates radiology findings (max 128 tokens default).
4. If XAI enabled: input-gradient saliency (Mac) or Grad-CAM (GPU) localizes attention.
5. Faithfulness module compares anatomy terms in text vs. heatmap focus region.
6. Results saved to outputs/ as JSON and PNG attribution overlays.

3. Technology Stack

Python 3.10+ | PyTorch | Hugging Face Transformers | OpenCV | Pillow
Model: BIOMEDICA/BMC-smolvlm1-256M (256M params, biomedical VLM)
Processor: HuggingFaceTB/SmolVLM-256M-Instruct
UI: Gradio 5+ (optional [ui] extra)
Runtime: Apple MPS / CUDA / CPU

4. Project Structure

```
meddiagnosis/  
  config/default.yaml          # Model, prompts, XAI settings  
  src/meddiagnosis/  
    cli.py                    # CLI entry (infer, serve)  
    config.py                 # YAML config loader  
    models/  
      smolvlm.py              # Local VLM wrapper  
      vlm_common.py           # Shared message/token helpers  
    pipeline/  
      inference.py            # DiagnosticPipeline orchestrator  
    xai/  
      gradcam.py              # Grad-CAM + fast saliency  
      visualization.py         # Heatmap overlay rendering  
    evaluation/  
      faithfulness.py          # Grounding score  
    app/  
      gradio_app.py           # Web UI + follow-up chat  
  outputs/  
    pyproject.toml            # Results and XAI images
```

5. Module Reference

config.py

Loads config/default.yaml. Exposes model_id, output_dir, ensure_dirs(), and Hugging Face token setup via HF_TOKEN environment variable.

models/vlm_common.py

Shared utilities: build_messages(), build_followup_messages(), prepare_inputs(), generate_from_messages(). Handles chat-template tokenization and device placement.

models/smolvlm.py - SmolVLMLocalModel

Wraps BIOMEDICA/BMC-smolvlm1-256M. Methods: generate(), chat() for follow-up Q&A, prepare_inputs() for XAI, get_vision_module() for attribution hooks. Auto-selects MPS/CUDA/CPU.

pipeline/inference.py - DiagnosticPipeline

Main orchestrator. Loads model + GradCAMExplainer. run() executes inference and optional XAI. Returns DiagnosticOutput (findings, gradcam_path, faithfulness). save_result() writes JSON.

xai/gradcam.py - GradCAMExplainer

Visual attribution over vision encoder. On Mac (MPS/CPU) uses fast input-gradient saliency (config xai.fast: true). On CUDA uses full Grad-CAM with hook-based activation gradients. Fallback to input-gradient on failure.

xai/visualization.py

normalize_heatmap(), overlay_heatmap() - blends JET colormap heatmap onto X-ray. save_attribution() writes PNG overlays.

evaluation/faithfulness.py

Extracts anatomy terms from generated text (lung regions, heart, pleura). Maps heatmap center-of-mass to anatomical bounding boxes. Computes binary faithfulness score: 1.0 if predicted focus region appears in explanation, else 0.0.

app/gradio_app.py

Gradio Blocks UI: image upload, prompt presets, Analyze button, Grad-CAM toggle, chatbot for follow-up questions on findings, faithfulness display. Preloads model on page load.

cli.py

Commands: infer (single-image diagnosis + XAI), serve (launch Gradio on port 7860).

6. Configuration (config/default.yaml)

```
model:
  local_id: BIOMEDICA/BMC-smolvlm1-256M
  processor_id: HuggingFaceTB/SmolVLM-256M-Instruct
  max_new_tokens: 128
  system_prompt: Expert radiologist assistant...
prompts:
  report_generation: Analyze this chest X-ray...
xai:
  output_dir: outputs/xai
```

```
alpha: 0.45      # overlay transparency
fast: true       # fast saliency on Mac
paths:
  outputs: outputs
```

7. Usage

Install

```
python -m venv .venv && source .venv/bin/activate
pip install -e ".[ui]"
```

CLI - Fast inference (no XAI)

```
python -m meddiagnosis.cli infer --image chest_xray.png --no-xai --max-tokens 80
```

CLI - With explainability (~2-3 min on Mac)

```
python -m meddiagnosis.cli infer --image chest_xray.png --max-tokens 80
```

Gradio Web UI

```
python -m meddiagnosis.cli serve --port 7860
# Open http://127.0.0.1:7860
```

8. Outputs

outputs/<sample-id>_result.json - findings, prompt, gradcam_path, faithfulness dict

outputs/xai/<sample-id>_gradcam.png - heatmap overlay on input X-ray

9. Research Contribution

Rather than training a foundation model from scratch, MedDiagnosis leverages a pre-trained biomedical VLM (BMC-SmolVLM1-256M) and augments it with a multi-level explainability framework:

- Visual attribution (Grad-CAM / input-gradient saliency)
- Language-level explanation (natural-language findings)
- Evidence grounding evaluation (anatomy term vs. heatmap region alignment)

This enables quantitative assessment of whether generated diagnostic explanations are faithful to the visual evidence used by the model.

10. Limitations

- Research prototype only - not validated for clinical use.
- BMC-smolVlm trained on general biomedical literature, not chest X-rays specifically.
- Faithfulness uses heuristic anatomy bounding boxes, not Chest Imagenome ground truth.
- XAI backward pass is slow on Apple MPS (~2-3 minutes).
- Model accuracy depends on prompt quality and image quality.